ENGINE CONTROL SYSTEM

1. General

The engine control system of the 2UZ-FE engine has the following system.

System	Outline
SFI (Sequential Multiport Fuel Injection)	• An L-type SFI system directly detects the intake air mass using a hot wire type air flow meter. • An independent injection system (in which fuel is injected once into each intake port for each two revolutions of the crankshaft) is used. • Fuel injection takes two forms: - Synchronous injection, in which injection always occurs at the same timing relative to the firing order. - The non-synchronous injection in which injection is effected regardless of the crankshaft angle.
ESA (Electronic Spark Advance)	• Ignition timing is determined by the ECM based on signals from various sensors. The ECM corrects ignition timing in response to engine knocking. • This system selects the optimal ignition timing in accordance with the signals received from the sensors and sends the (IGT) ignition signal to the igniter.
ETCS-i (Electronic Throttle Control System-intelligent) [See page EG-116]	Optimally controls the throttle valve opening in accordance with the amount of accelerator pedal effort and the condition of the engine and the vehicle.
VVT-i (Variable Valve Timing-intelligent) [See page EG-119]	Controls the intake camshaft to optimal valve timing in accordance with the engine condition.
ACIS (Acoustic Control Induction System) [See page EG-123]	The intake air passages are switched according to the engine speed and throttle valve opening angle to provided high performance in all speed ranges.
Fuel Pump Control	• Based on signals from the ECM, the fuel pump ECU controls the fuel pump to 3 stages. • The fuel pump is stopped when the SRS airbag is deployed in a frontal, side, or side rear collision. • The basic construction and operation of the fuel pump control are the same as in the 1GR-FE engine. For details, see page EG-68.
Air Injection Control* [See page EG-125]	The ECM controls the air injection time based on the signals from the crankshaft position sensor, engine coolant temp. sensor, mass air flow meter and air pressure sensor.
Starter Control (Cranking Hold Function)	• Once the engine switch is pushed, this control continues to operate the starter until the engine started. • The basic construction and operation of the starter control are the same as in the 1GR-FE engine. For details, see page EG-70.
Air Fuel Ratio Sensor and Oxygen Sensor Heater Control	Maintains the temperature of the air fuel ratio sensors or oxygen sensors at an appropriate level to increase the detection accuracy of the exhaust gas oxygen concentration.

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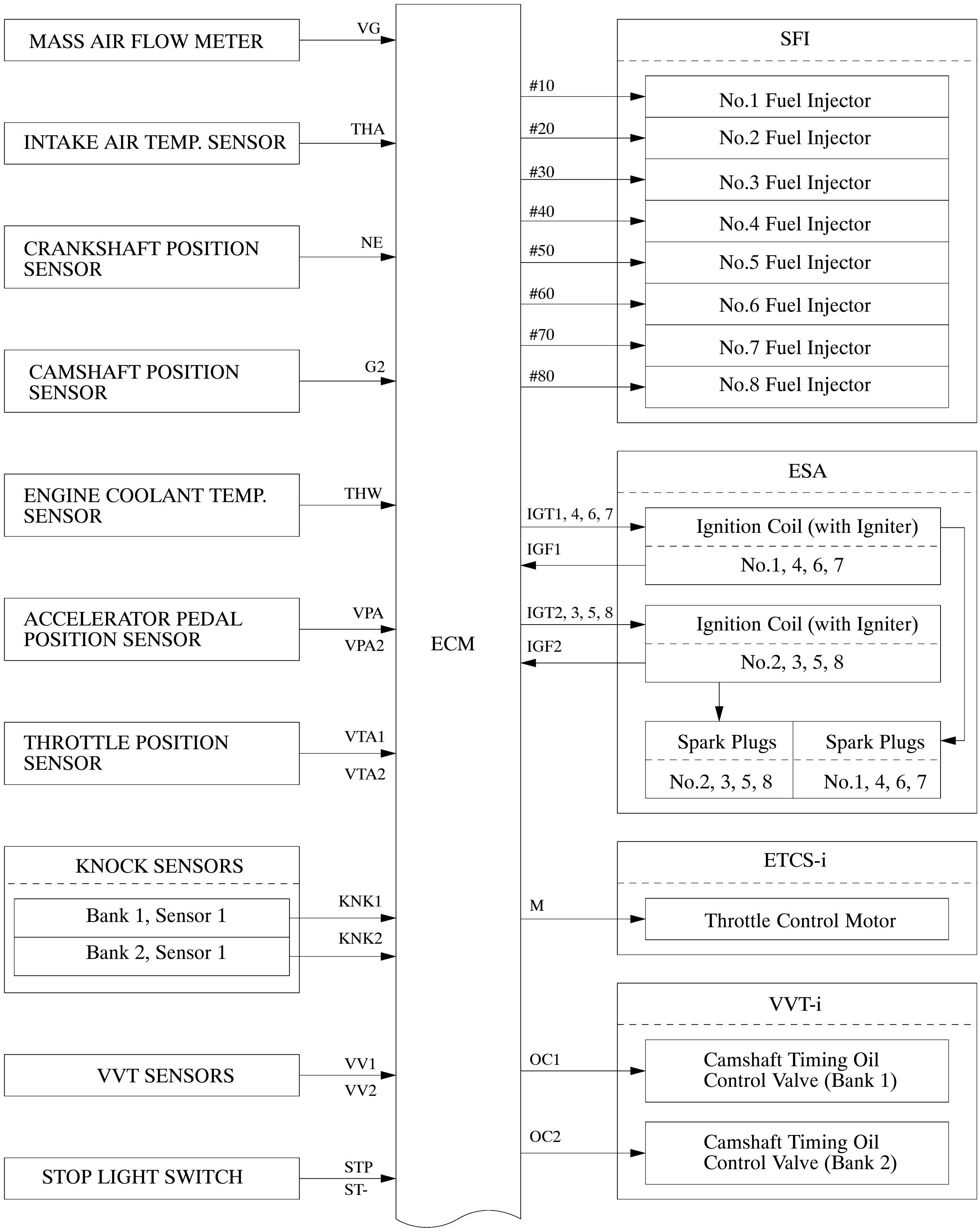
System	Outline
Air Conditioning Cut-off Control	By turning the air conditioning compressor ON or OFF in accordance with the engine condition, drivability is maintained.
Evaporative Emission Control	The ECM controls the purge flow of evaporative emission (HC) in the charcoal canister in accordance with engine conditions.
Engine Immobilizer	Prohibits fuel delivery and ignition if an attempt is made to start the engine with an invalid key.
Diagnosis [See page EG-130]	When the ECM detects a malfunction, the ECM records the malfunction and memorizes information related to the fault.
Fail-safe [See page EG-130]	When the ECM detects a malfunction, the ECM stops or controls the engine according to the data already stored in the memory.

*: Only for European, Australian and Chinese models

2. Construction

The configuration of the engine control system in the 2UZ-FE engine is as shown in the following chart.



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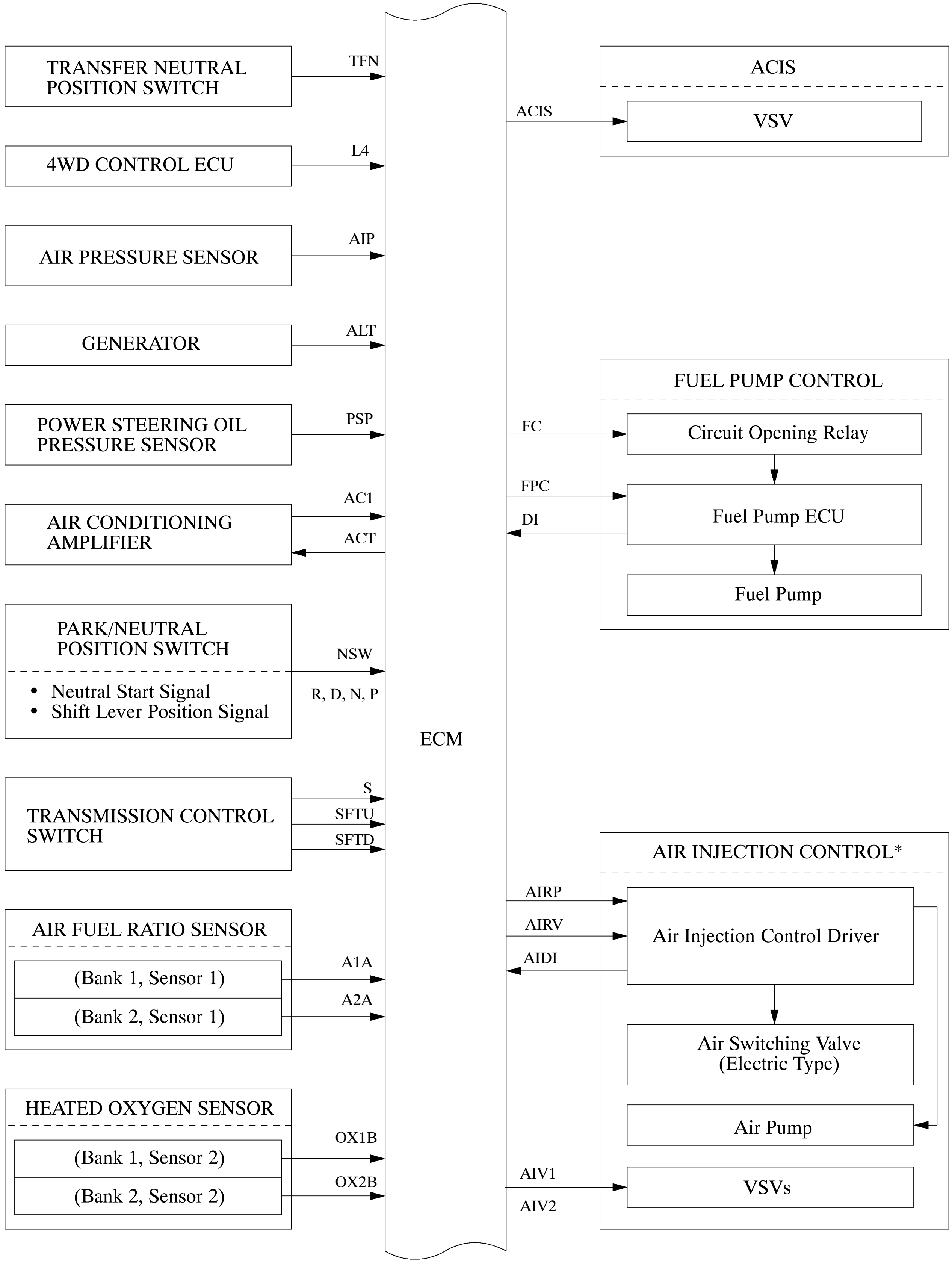
EG


TRANSFER NEUTRAL POSITION SWITCH
TFN
4WD CONTROL ECU
L4
AIR PRESSURE SENSOR
AIP
GENERATOR
ALT
POWER STEERING OIL PRESSURE SENSOR
PSP
AIR CONDITIONING AMPLIFIER
AC1
ACT
PARK/NEUTRAL POSITION SWITCH
• Neutral Start Signal
• Shift Lever Position Signal
NSW
R, D, N, P
TRANSMISSION CONTROL SWITCH
S
SFTU
SFTD
AIR FUEL RATIO SENSOR
(Bank 1, Sensor 1)
(Bank 2, Sensor 1)
A1A
A2A
HEATED OXYGEN SENSOR
(Bank 1, Sensor 2)
(Bank 2, Sensor 2)
OX1B
OX2B
ECM
ACIS
ACIS
VSV
FUEL PUMP CONTROL
FC
Circuit Opening Relay
FPC
DI
Fuel Pump ECU
Fuel Pump
AIR INJECTION CONTROL*
AIRP
AIRV
AIDI
Air Injection Control Driver
Air Switching Valve (Electric Type)
Air Pump
AIV1
AIV2
VSVs
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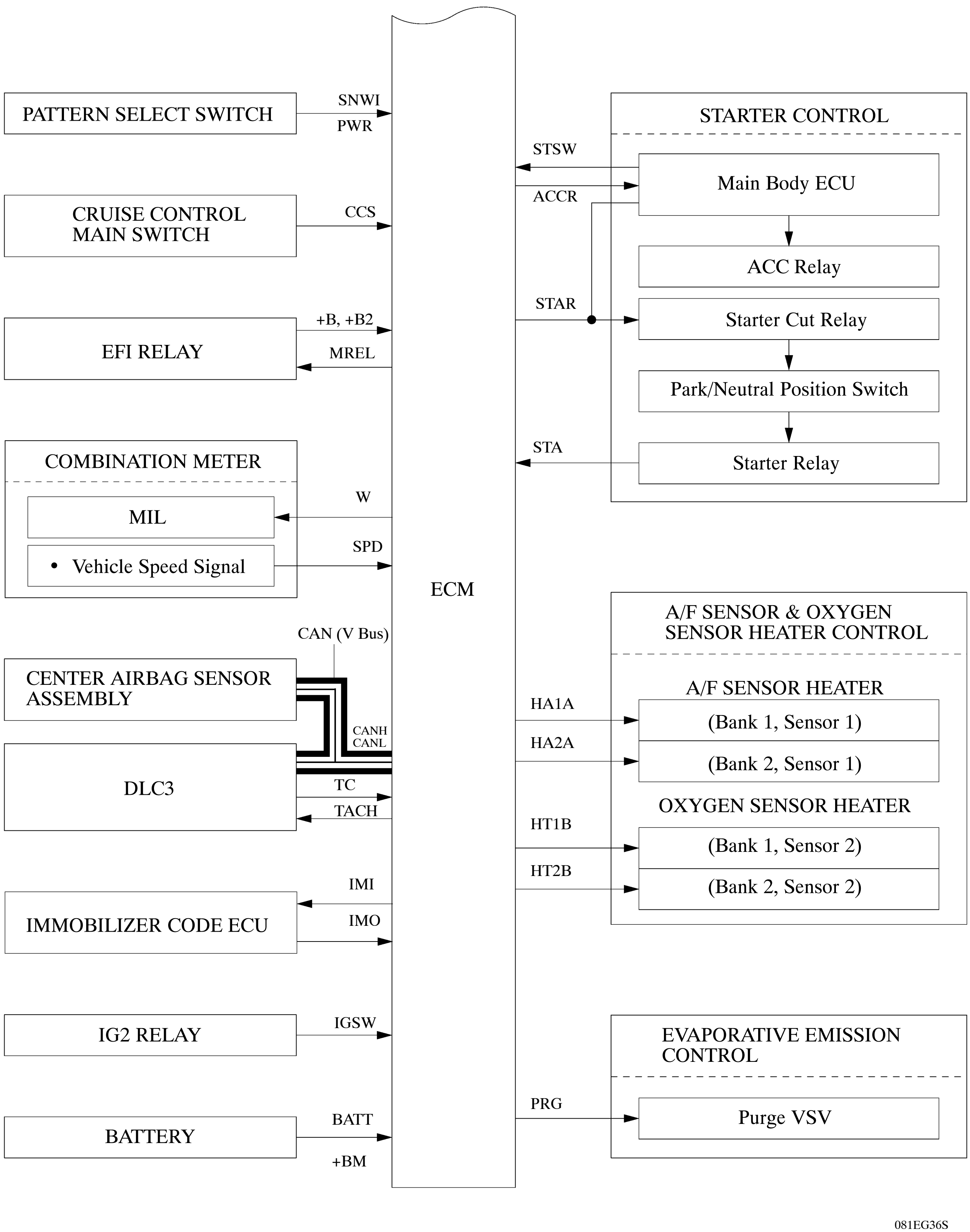
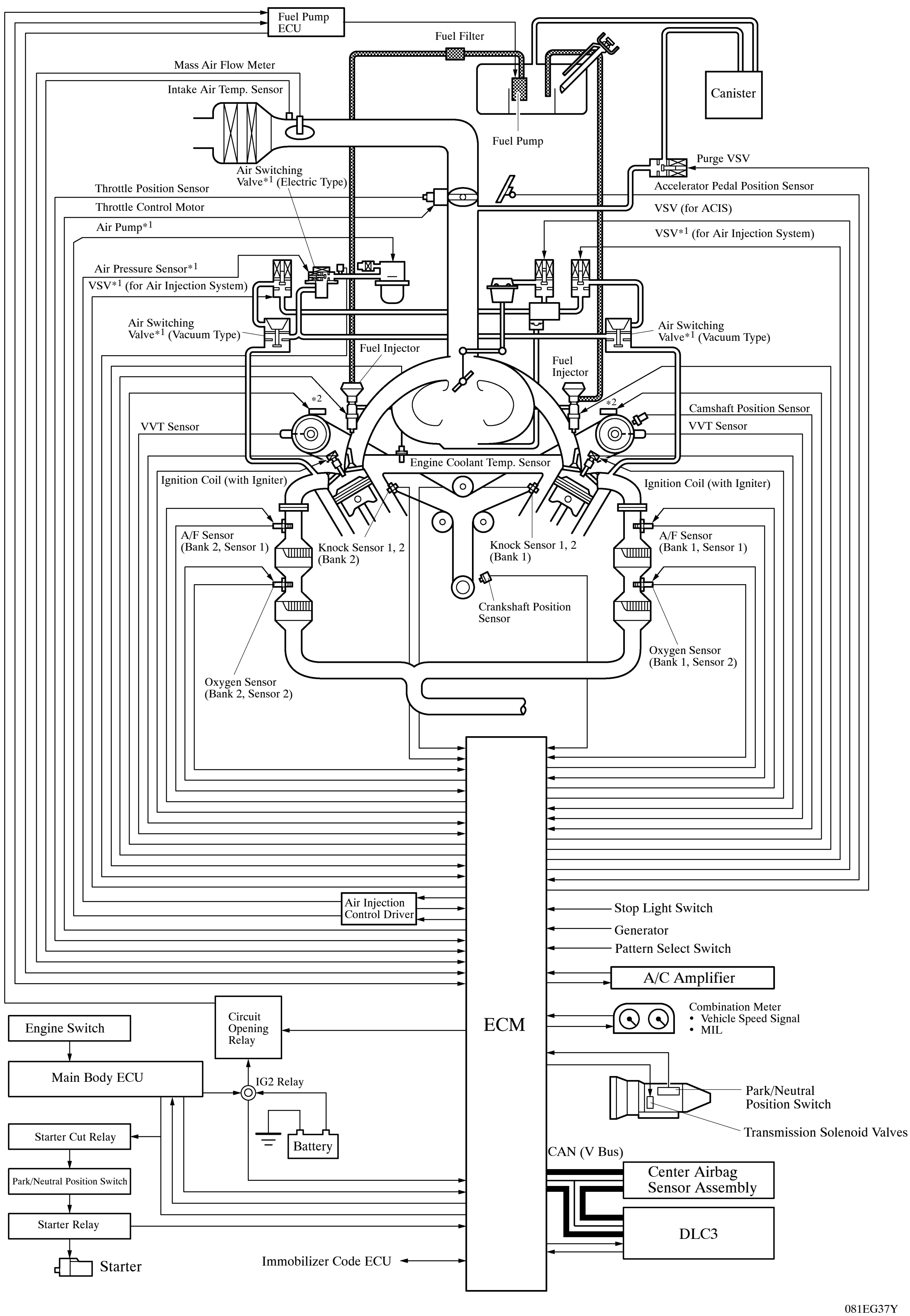


*: Only for European, Australian and Chinese models

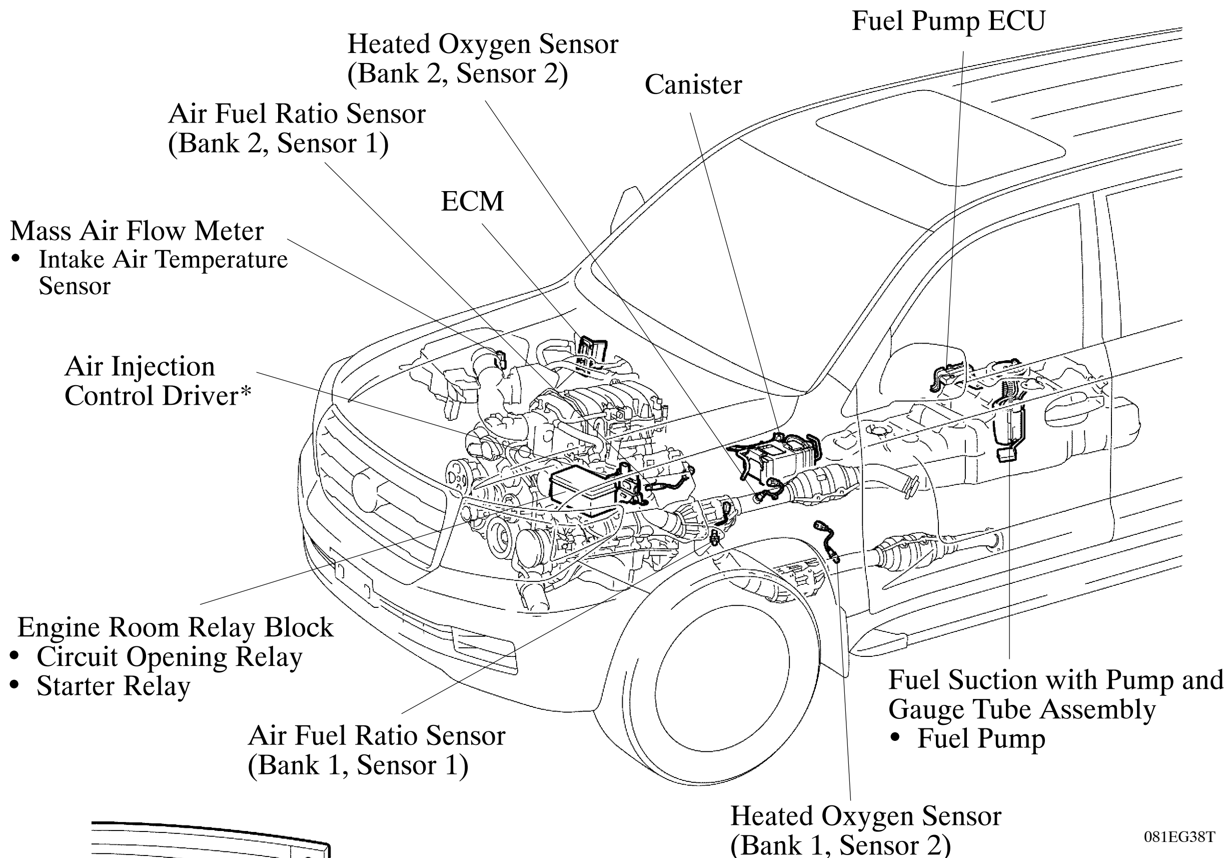
EG

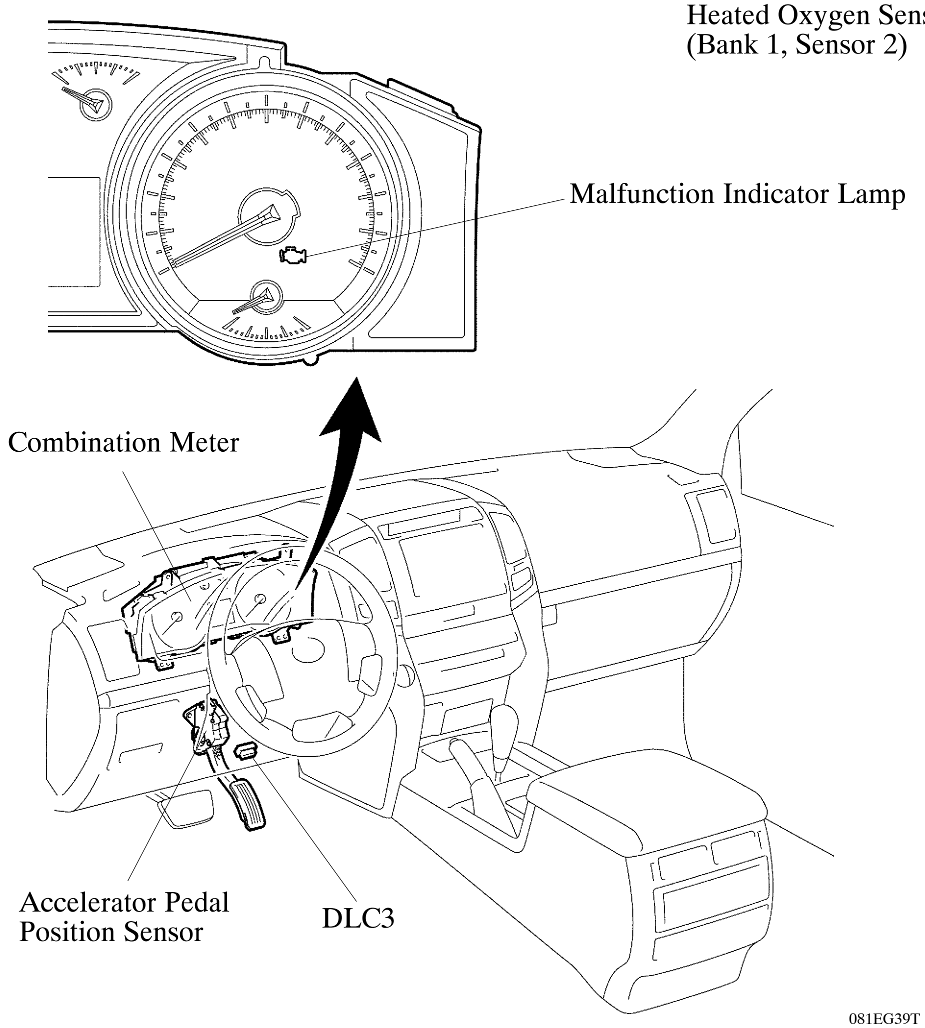
3. Engine Control System Diagram



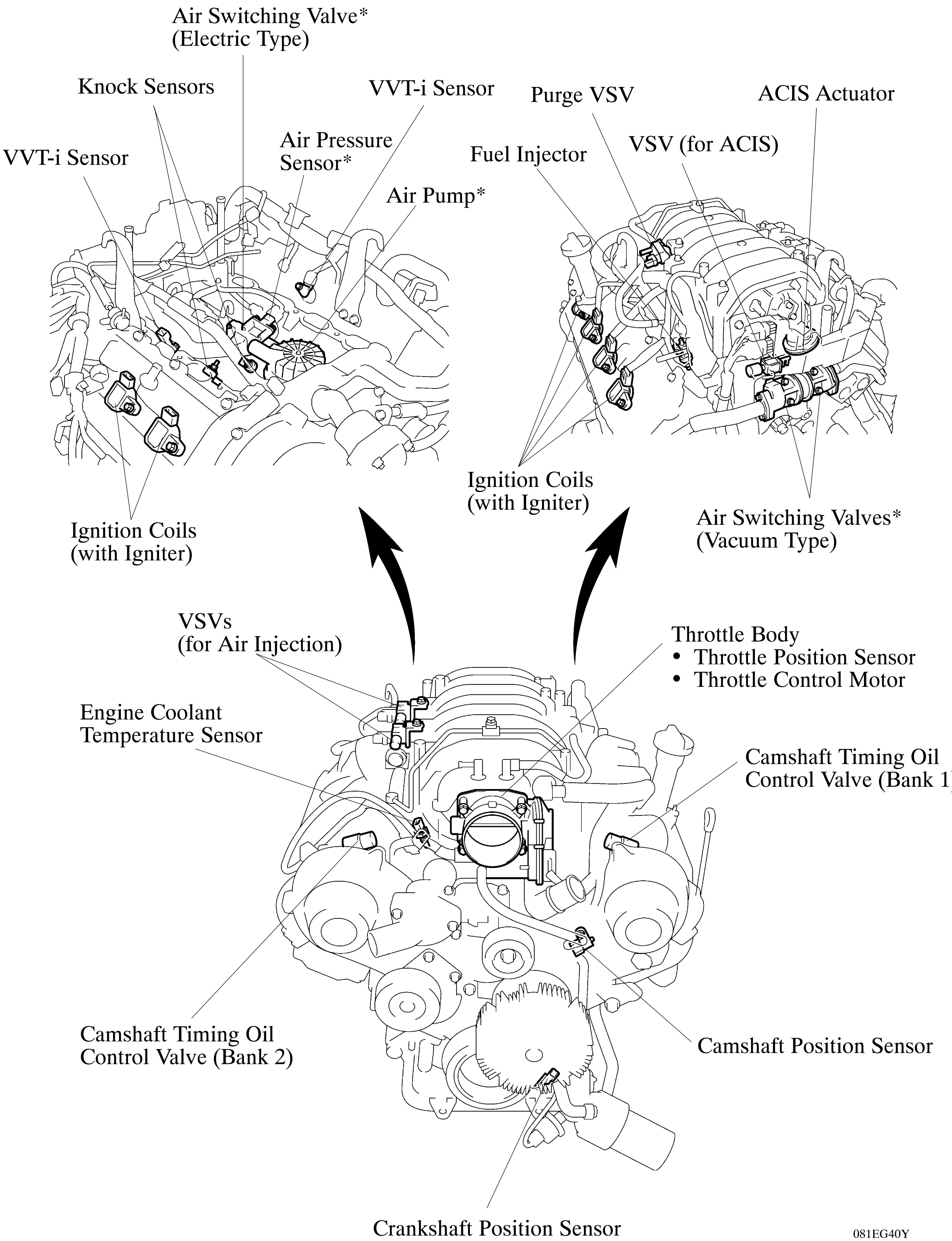
*1: Only for European, Australian and Chinese models
*2: Camshaft Timing Oil Control Valve

4. Layout of Main Components





*: Only for European, Australian and Chinese models



*: Only for European, Australian and Chinese models

5. Main Components of Engine Control System

General

The main components of the 2UZ-FE engine control system are as follows:

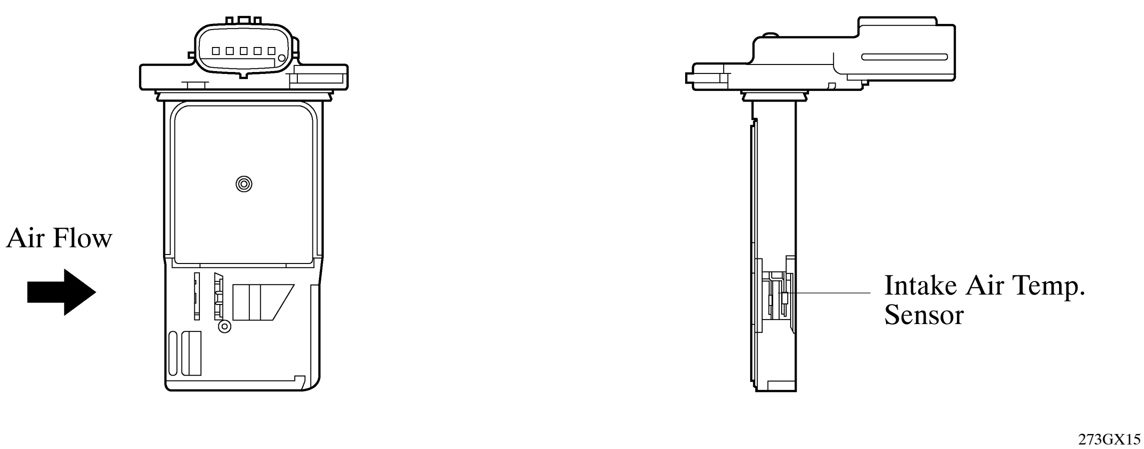
Components	Outline	Quantity	Function
ECM	32-bit CPU (DENSO)	1	The ECM optimally controls the SFI, ESA and ISC to suit the operating conditions of the engine in accordance with the signals provided by the sensors.
Mass Air Flow Meter [See page EG-112]	Hot-wire Type	1	This sensor has a built-in hot-wire to directly detect the intake air mass and flow rate.
Intake Air Temperature Sensor	Thermistor Type	1	This sensor detects the intake air temperature by means of an internal thermistor.
Crankshaft Position Sensor [See page EG-113]	Pick-up Coil Type (Rotor Teeth /36-2)	1	This sensor detects the engine speed and the crankshaft position.
Camshaft Position Sensor [See page EG-113]	MRE Type (Rotor Teeth /3)	1	This sensor detects the camshaft position and performs the cylinder identification.
VVT Sensor [See page EG-115]	Pick-up Coil Type (Rotor Teeth /3)	1 each bank	This sensor detects the actual valve timing.
Accelerator Pedal Position Sensor	No-contact Type	1	• This sensor detects the amount of pedal effort applied to the accelerator pedal. • The basic construction and operation of this sensor are the same as in the 1GR-FE engine. For details, see page EG-54.
Throttle Position Sensor	No-contact Type	1	• This sensor detects the throttle valve opening angle. • The basic construction and operation of this sensor are the same as in the 1GR-FE engine. For details, see page EG-55.
Knock Sensor	Built-in Piezoelectric Type (Flat Type)	1 each bank	• This sensor detects an occurrence of the engine knocking indirectly from the vibration of the cylinder block caused by the occurrence of engine knocking. • The basic construction and operation of this sensor are the same as in the 1GR-FE engine. For details, see page EG-56.
Heated Oxygen Sensor	Cup Type with Heater	1 each bank	• This sensor detects the oxygen concentration in the exhaust gas by measuring the electromotive force which is generated in the sensor itself. • The basic construction and operation of this sensor are the same as in the 1GR-FE engine. For details, see page EG-58.

(Continued)

Components	Outline	Quantity	Function
Air Fuel Ratio Sensor	Planar Type with Heater	1 each bank	• As with the oxygen sensor, this sensor detects the oxygen concentration in the exhaust gas. However, it detects the oxygen concentration in the exhaust gas linearly. • The basic construction and operation of this sensor are the same as in the 1GR-FE engine. For details, see page EG-58.
Engine Coolant Temperature Sensor	Thermistor Type	1	This sensor detects the engine coolant temperature by means of an internal thermistor.
Fuel Injector [See page EG-96]	4-Hole Type	8	This fuel injector contains an electro-magnetically operated nozzle to inject fuel into the intake port.
Camshaft Timing Oil Control Valve [See page EG-121]	Electro-Magnetic Coil Type	2 each bank	The camshaft timing oil control valve changes the valve timing by switching the oil passage that acts on the VVT-i controller in accordance with the signals received from the ECM.

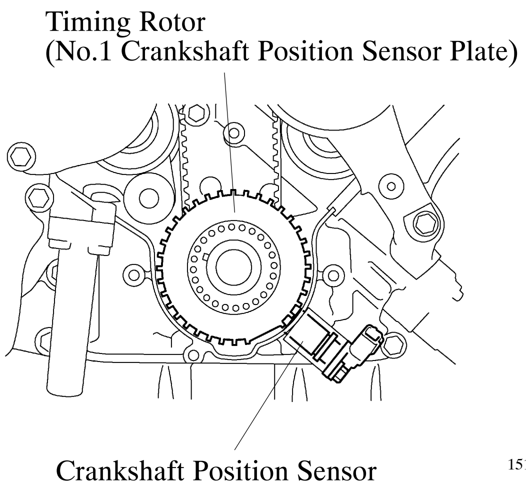
Mass Air Flow Meter

- This mass air flow meter, which is a slot-in type, allows a portion of the intake air to flow through the detection area. By directly measuring the mass and the flow rate of the intake air, the detection precision is improved and the intake air resistance is reduced.
- This mass air flow meter has a built-in intake air temperature sensor.



Crankshaft Position Sensor

The timing rotor of the crankshaft consists of a 34 tooth plate with 2 teeth missing. The crankshaft position sensor outputs a crankshaft rotation signal every 10° of crankshaft rotation, and the change of the signal due to the missing teeth is used to determine top-dead-center.



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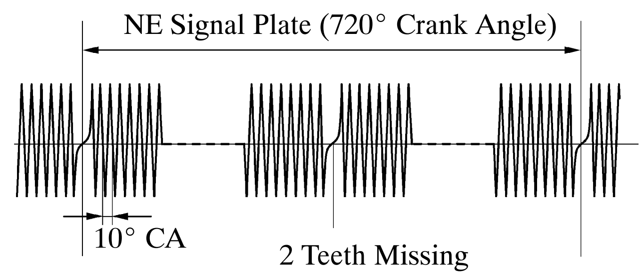

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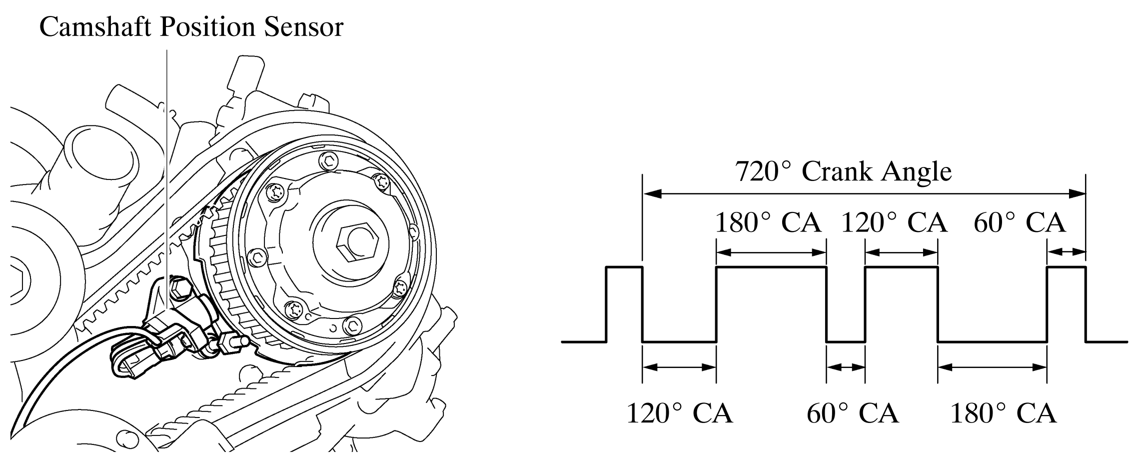
Camshaft Position Sensor

1) General

The camshaft position sensor uses a timing rotor that is provided on the VVT-i controller of the left bank. Based on the timing rotor, the sensor outputs camshaft position signals consisting of 3 (3 Hi output, 3 Lo output) pulses for every 2 revolutions of the crankshaft.



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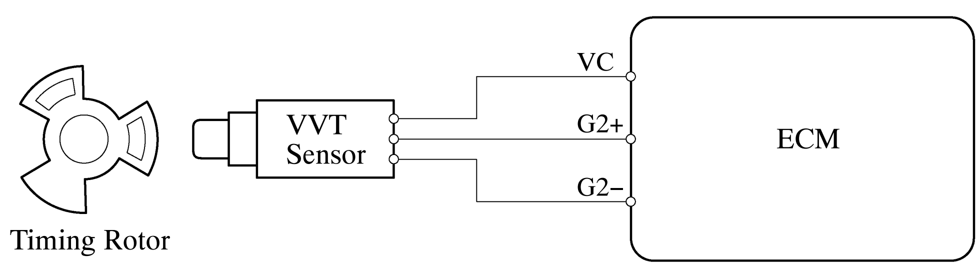
2) Construction and Operation

- The MRE type camshaft position sensor consists of an MRE, a magnet and a sensor. The direction of the magnetic field changes due to the different shapes (protruded and non-protruded portions) of the timing rotor, which passes by the sensor. As a result, the resistance of the MRE changes, and the output voltage to the ECM changes to Hi or Lo. The ECM detects the camshaft position based on this output voltage. The ECM compares the camshaft position sensor signal and crankshaft position sensor signals to detect the camshaft position and identify the cylinder.
- The differences between the MRE type camshaft position sensor and the pickup coil type camshaft position sensor used on the conventional model are as follows.

Item	Sensor Type	
	MRE	Pick up Coil
Signal Output	Constant digital output starts from low engine speeds	Analog output changes with the engine speed
Camshaft Position Detection	Detection based on the waveforms output throughout the timing rotor speed range	Detection based on the waveforms output as the protruded portion of the timing rotor passes

▶ Wiring Diagram ◀



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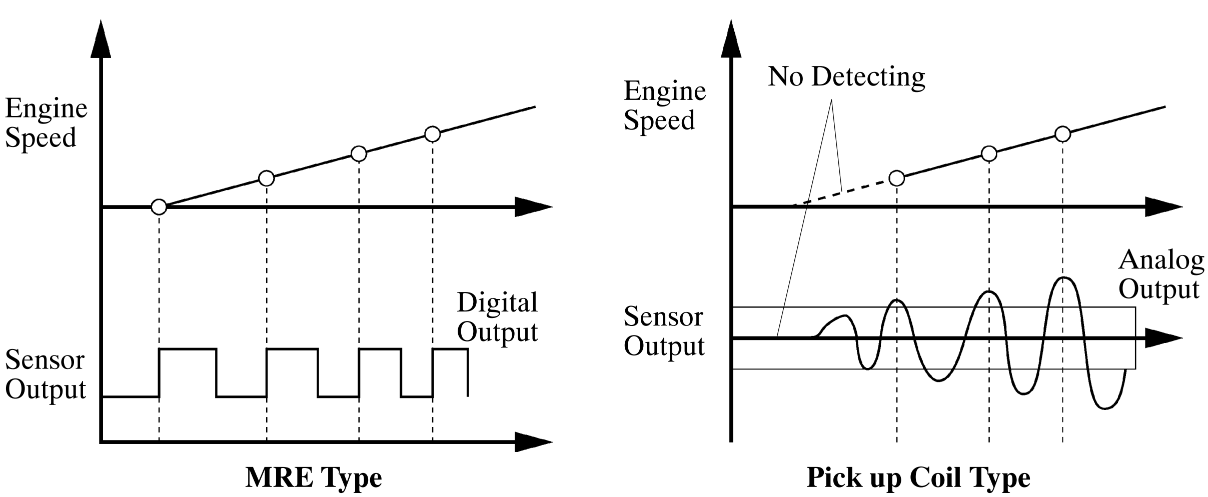
▶ MRE Type and Pickup Coil Type Output Waveform Image Comparison ◀



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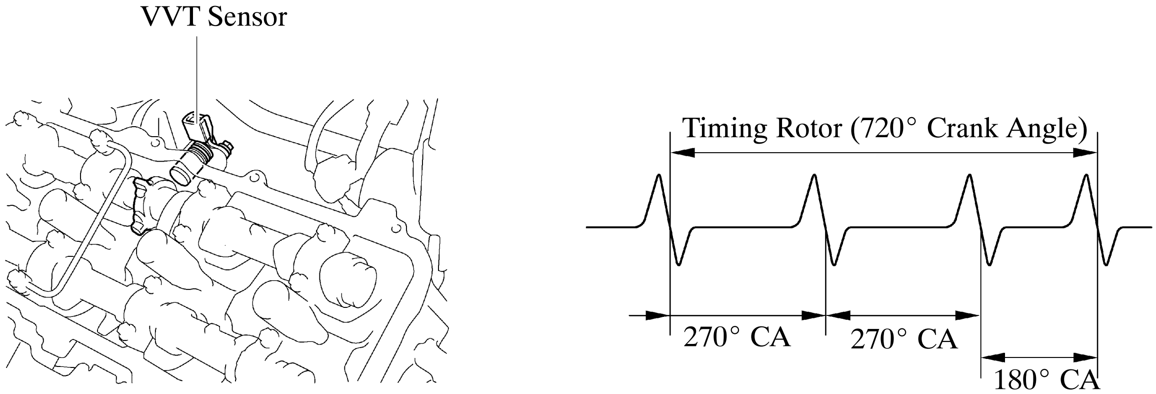
VVT Sensor

A VVT sensor is mounted on the intake side of each cylinder head. To detect the camshaft position, a timing rotor that is provided on the intake camshaft is used to generate 3 pulses for every 2 revolutions of the crankshaft. The ECM compares these VVT sensor signals and the crankshaft position sensor signal to detect the actual valve timing.



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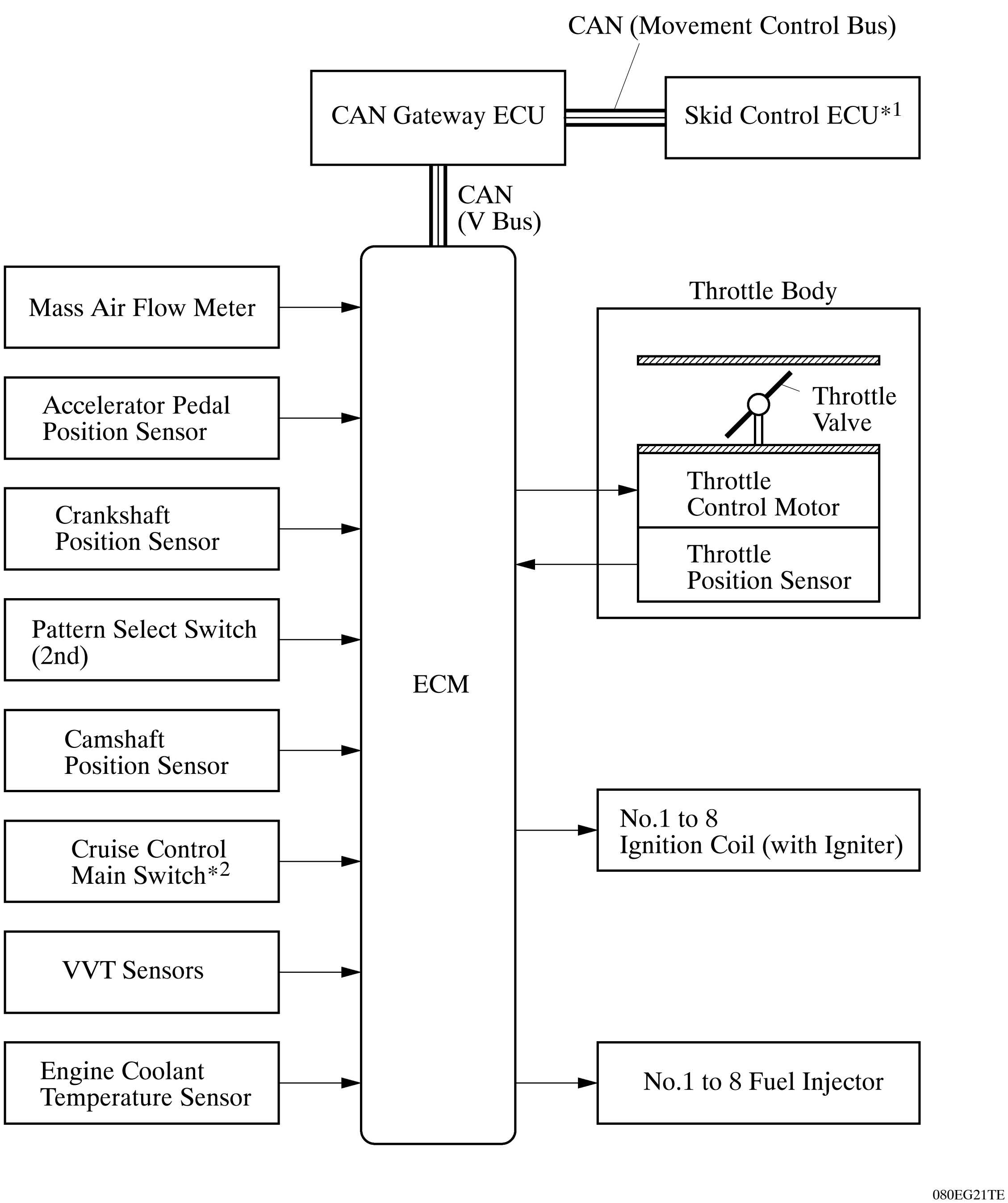
6. ETCS-i (Electronic Throttle Control System-intelligent)

General

- In the conventional throttle body, the throttle valve angle is determined invariably by the amount of the accelerator pedal effort. In contrast, ETCS-i uses the ECM to calculate the optimal throttle valve angle that is appropriate for the respective driving condition and uses a throttle control motor to control the angle.
- In case of an abnormal condition, this system transfers to the fail-safe mode. For details, see page EG-131.

▶ System Diagram ◀



*1: Only for models with brake control system (Multi-terrain ABS with EBD, brake assist, A-TRC, VSC, hill-start assist control and CRAWL)
*2: Only for models with cruise control system

Control

1) General

The ETCS-i consists of the following functions:

- Normal Throttle Control (non-linear control)
- ISC (Idle Speed Control)
- A-TRC (Active Traction Control)*1
- VSC (Vehicle Stability Control) Coordination Control*1
- CRAWL (Crawl Control) Coordination Control*1
- Cruise Control*2

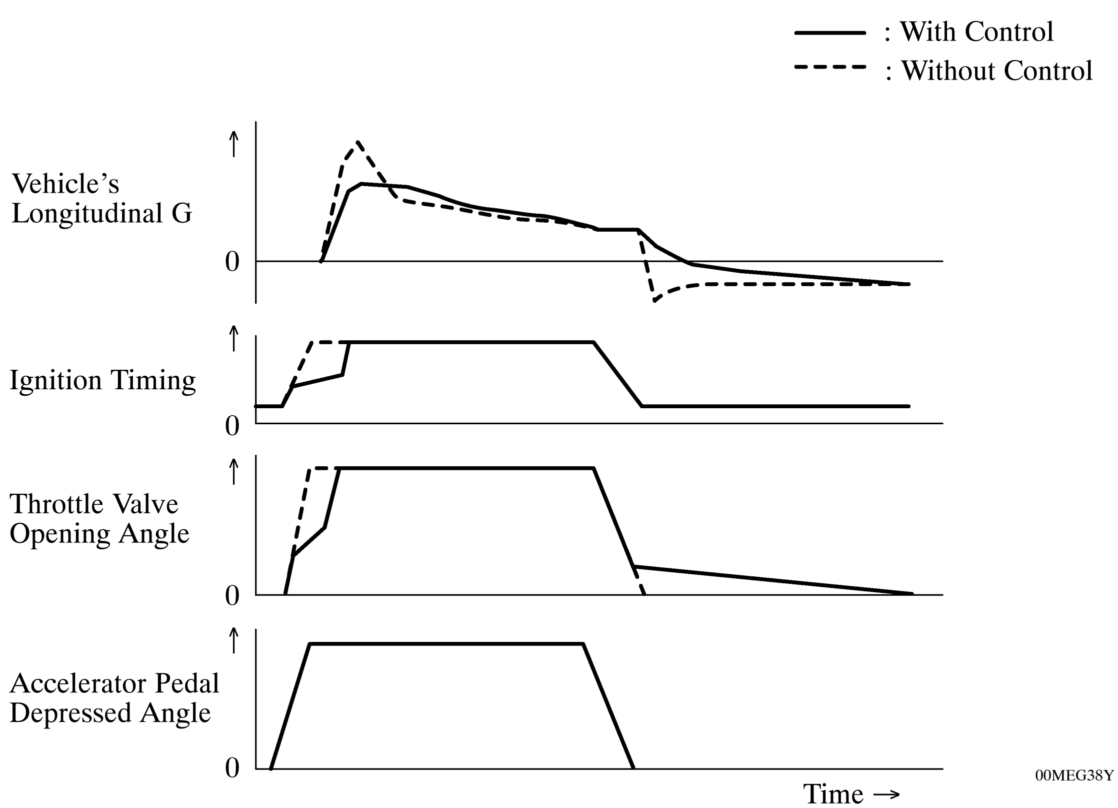
*1: Only for models with brake control system (Multi-terrain ABS with EBD, brake assist, A-TRC, VSC, hill-start assist control and CRAWL)
*2: Only for models with cruise control system

2) Normal Throttle Control (Non-linear Control)

a. General

Controls the throttle to an optimal throttle valve angle that is appropriate for the driving condition such as the amount of the accelerator pedal effort and the engine speed in order to realize excellent throttle control and comfort in all operating ranges.

▶ Conceptual Diagrams of Engine Control During Acceleration and Deceleration ◀



b. 2nd Start Control

In situations in which low-μ (low friction) road surface conditions can be anticipated, such as when driving in the snow, the rate of throttle valve opening can be controlled to help vehicle stability while driving on the slippery surface. This is accomplished by turning on 2nd start control. Pressing the 2nd side of the pattern select switch activates this control. This control modifies the relationship and reaction of the throttle to the accelerator pedal, and assists the driver by reducing the engine output from that of a normal level.

3) Idle Speed Control

The ECM controls the throttle valve in order to constantly maintain an ideal idle speed.

4) A-TRC

As part of the A-TRC the throttle valve opening is reduced by a demand signal sent from the skid control ECU to the ECM. This demand signal will be sent if an excessive amount of slippage occurs at a drive wheel, thus facilitating vehicle stability and the application of an appropriate amount of power to the road.

5) VSC Coordination Control

In order to bring the effectiveness of the VSC into full play, the throttle valve angle is regulated through a coordination control with the skid control ECU.

6) CRAWL Coordination Control

While the CRAWL is operating, the throttle valve is regulated through a coordination control with the skid control ECU.

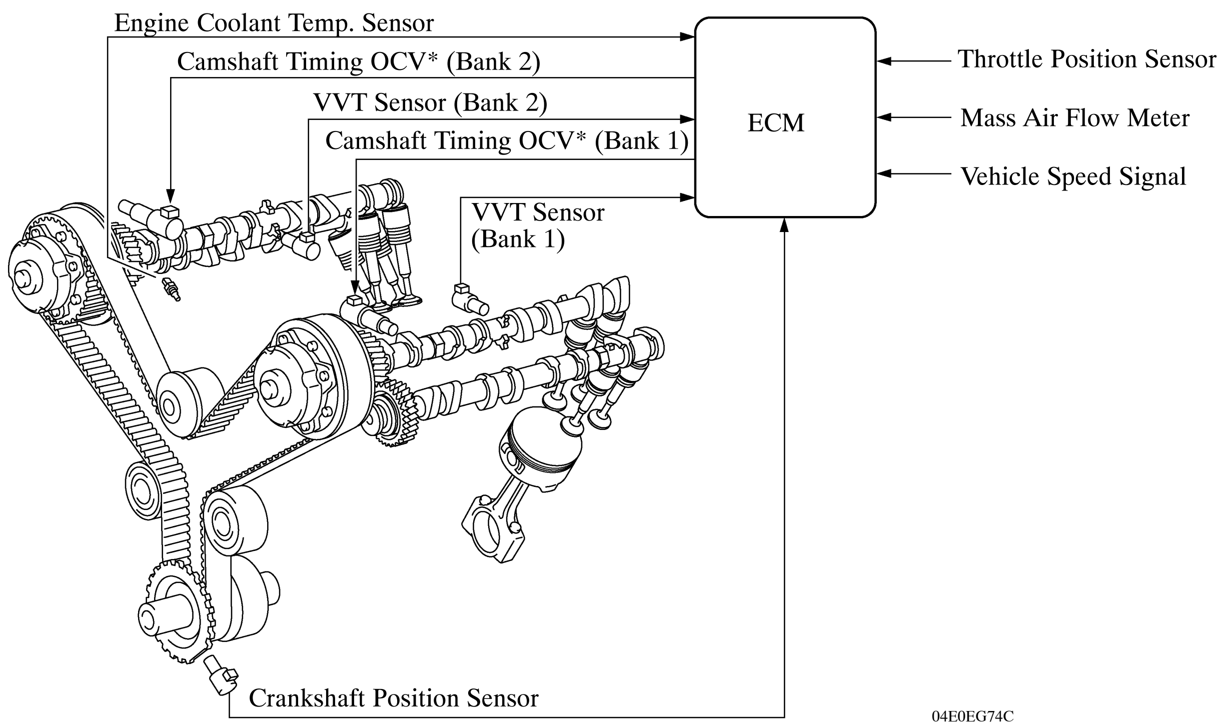
7) Cruise Control

The ECM directly actuates the throttle valve for operation of the cruise control.

7. VVT-i (Variable Valve Timing-intelligent) System

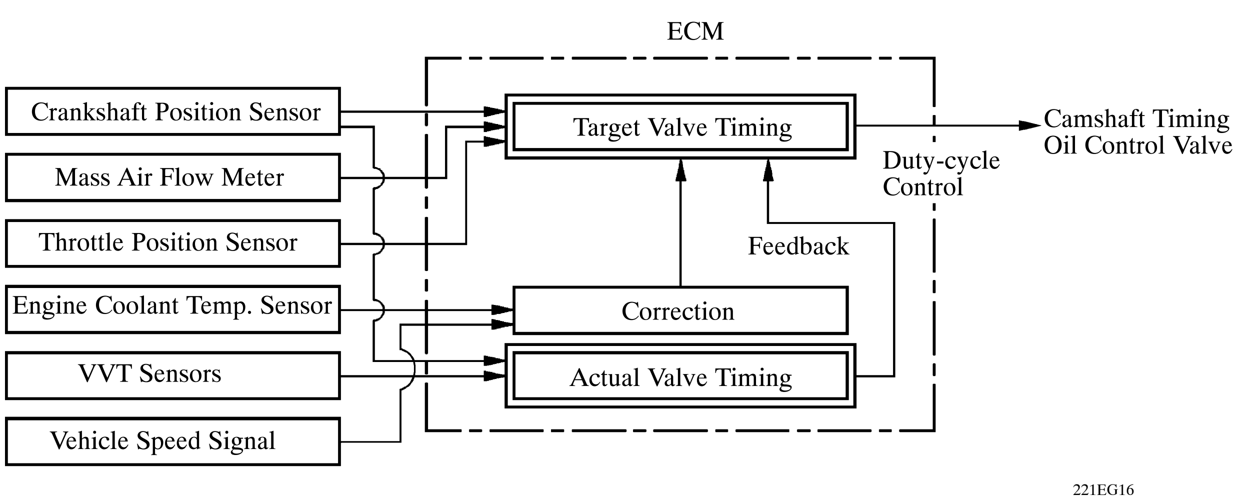
General

- The VVT-i system is designed to control the intake camshaft within a range of 40° (of Crankshaft Angle) to provide valve timing that is optimally suited to the engine condition. This improves torque in all the speed ranges as well as increasing fuel economy, and reducing exhaust emissions.



*: Oil Control Valve

- By using the engine speed, intake air mass, throttle position, engine coolant temperature and vehicle speed, the ECM can calculate optimal valve timing for each driving condition and controls the camshaft timing oil control valve. In addition, the ECM uses signals from the VVT sensor and the crankshaft position sensor to detect the actual valve timing, thus providing feedback control to achieve the target valve timing.



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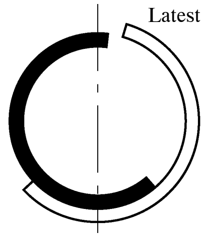
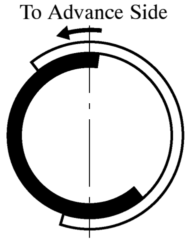
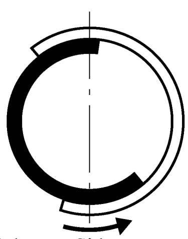
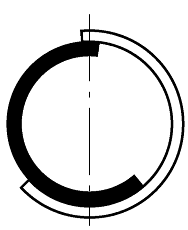
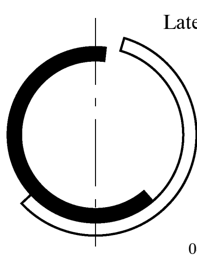
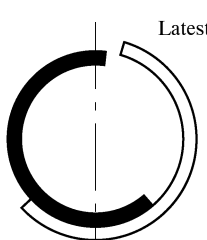
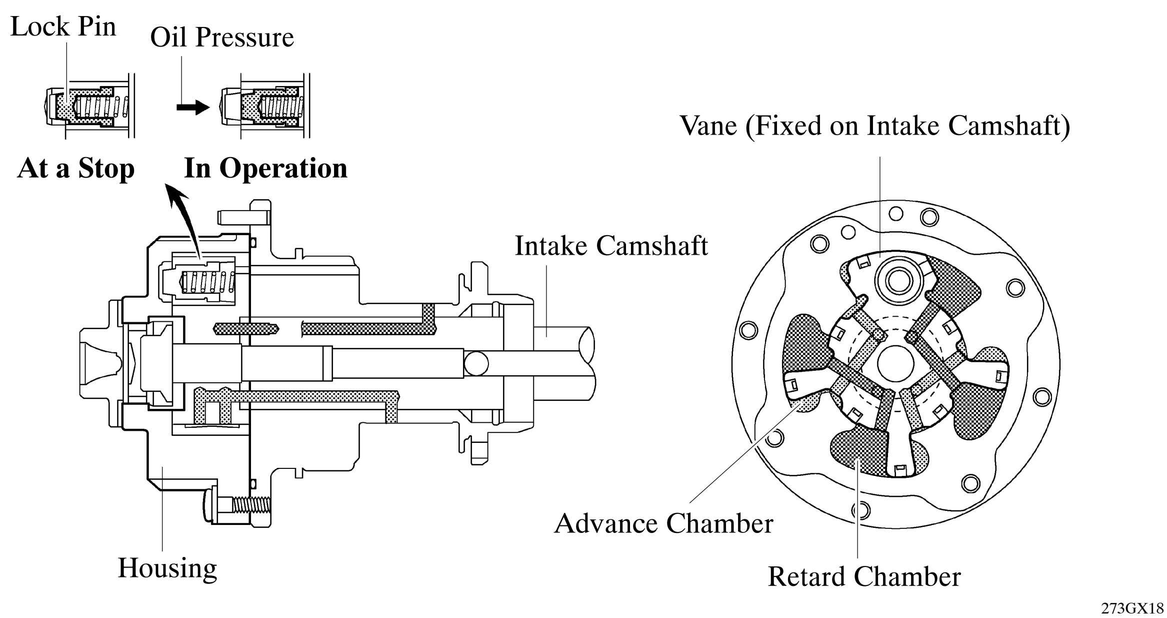
Effectiveness of the VVT-i System

Operation State	Objective		Effect
• During Idling • At Light Load		Eliminating overlap to reduce blow back to the intake side	• Stabilized idling rpm • Better fuel economy
At Medium Load		Increasing overlap to increase internal EGR to reduce pumping loss	• Better fuel economy • Improved emission control
In Low to Medium Speed Range with Heavy Load		Advancing the intake valve close timing for volumetric efficiency improvement	Improved torque in low to medium speed range
In High Speed Range with Heavy Load		Retarding the intake valve close timing for volumetric efficiency improvement	Improved output
At Low Temperatures		Eliminating overlap to prevent blow back to the intake side and stabilizes the idling speed at fast idle.	• Stabilized fast idle rpm • Better fuel economy
• Upon Starting • Stopping the Engine		Eliminating overlap to reduce blow back to the intake side	Improved startability

Construction

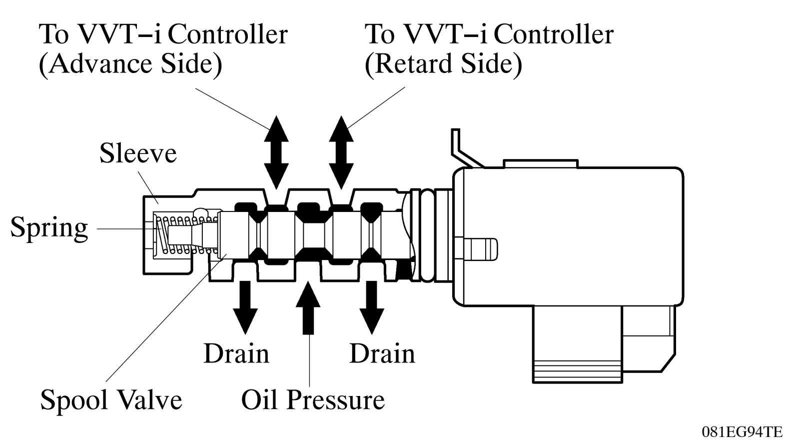
1) VVT-i Controller

VVT-i controller consists of the housing, four vanes, and a lock pin. The oil pressure sent from the advance or retard side path of the intake camshaft causes rotation in the VVT-i controller vane circumferential direction to vary the intake valve timing continuously. When the engine is stopped, the intake camshaft will be in the most retarded state to ensure startability. When hydraulic pressure is not applied to the VVT-i controller immediately after the engine has started, the lock pin locks the movement of the VVT-i controller to prevent a knocking noise. Thereafter, when hydraulic pressure is applied to the VVT-i controller, the lock pin is released.



2) Camshaft Timing Oil Control Valve

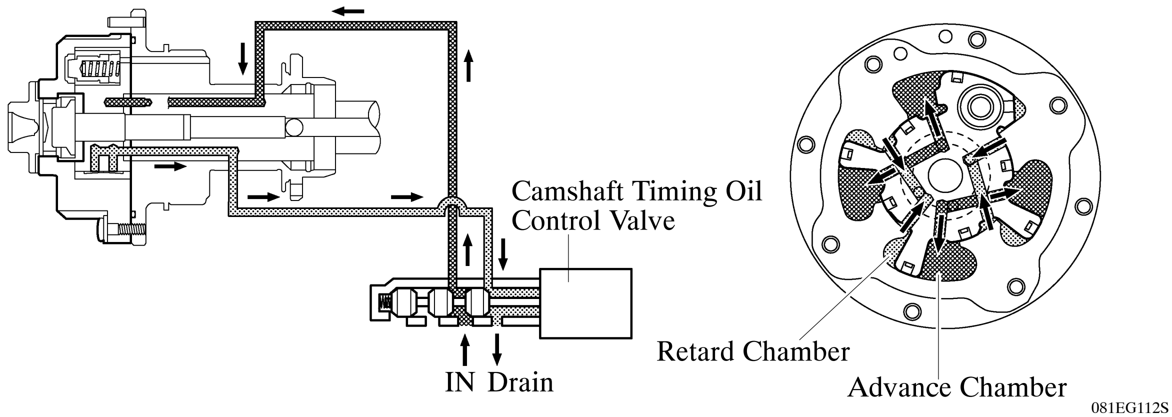
This camshaft timing oil control valve controls the spool valve using duty-cycle control from the ECM. This allows hydraulic pressure to be applied to the VVT-i controller advance or retard side. When the engine is stopped, the camshaft timing oil control valve is in the most retard position.



Operation

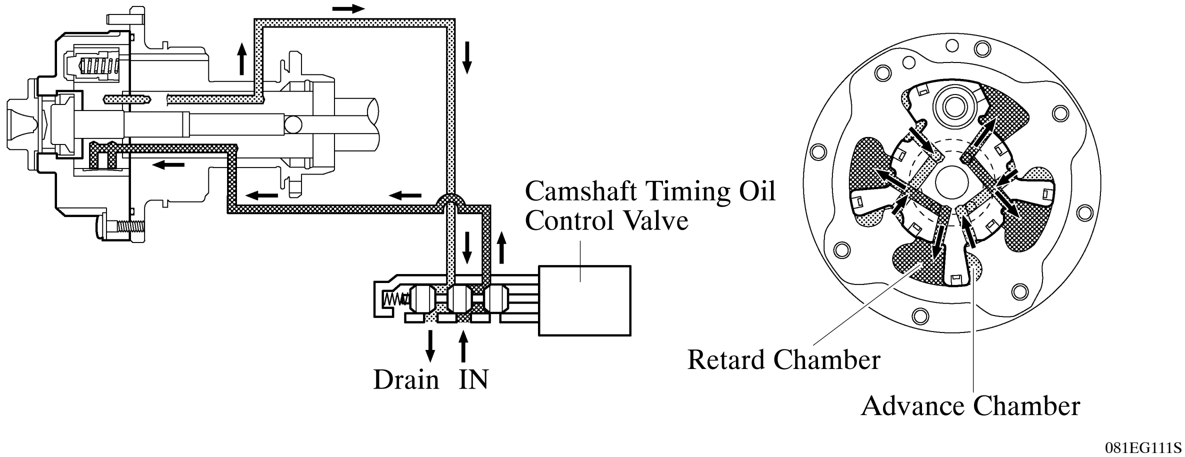
1) Advance

When the camshaft timing oil control valve is positioned as illustrated below by the advance signals from the ECM, the resultant oil pressure is applied to the timing advance side vane chamber to rotate the camshaft in the timing advance direction.



2) Retard

When the camshaft timing oil control valve is positioned as illustrated below by the retard signals from the ECM, the resultant oil pressure is applied to the timing retard side vane chamber to rotate the camshaft in the timing retard direction.



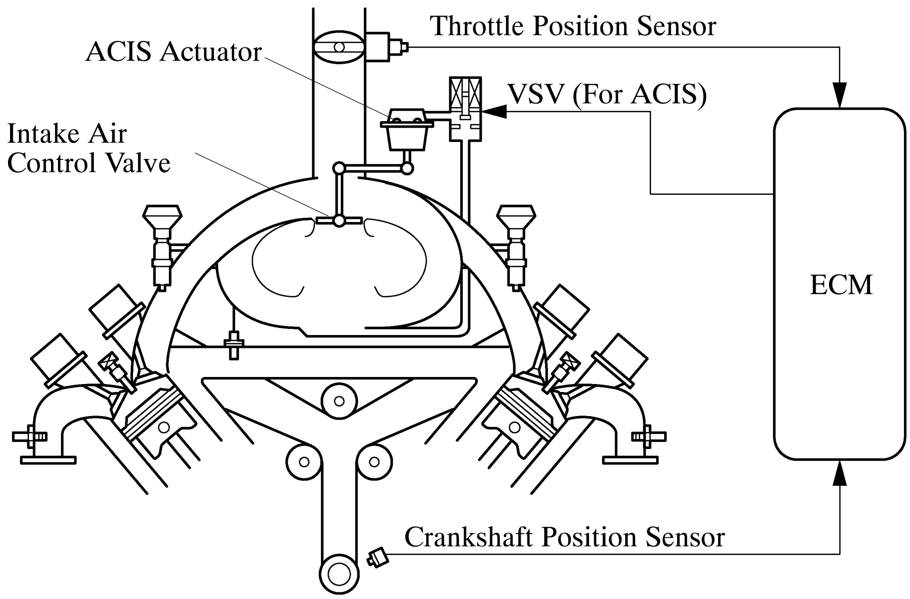
3) Hold

After reaching the target timing, the valve timing is held by keeping the camshaft timing oil control valve in the neutral position unless the traveling state changes.
This adjusts the valve timing at the desired target position and prevents the engine oil from running out when it is unnecessary.

8. ACIS (Acoustic Control Induction System)

General

The ACIS uses the intake air control valve as a bulkhead to divide the intake manifold into 2 stages. The intake air control valve is opened and closed to vary the effective length of the intake manifold in accordance with engine speed and throttle valve opening angle. This increases the power output in all ranges from low to high engine speeds.



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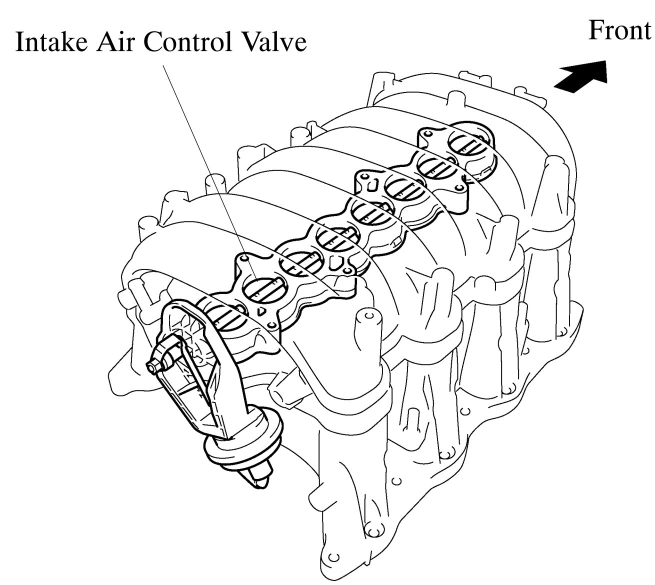
Intake Air Control Valve and Actuator

- The intake air control valve and actuator are integrated with the intake manifold.
- The intake air control valve opens and closes to make two effective lengths of the intake manifold possible.



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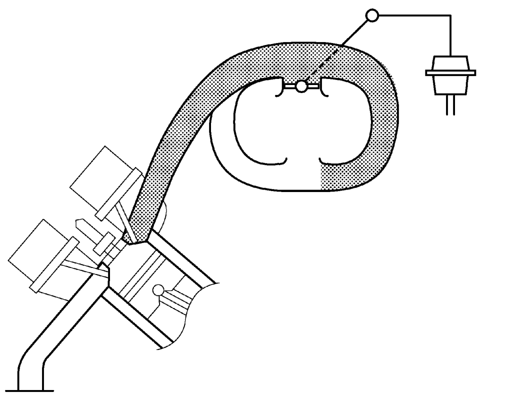
Operation

1) Intake Air Control Valve Close (VSV ON)

The ECM activates the VSV to match the longer pulsation cycle so that the negative pressure acts on the diaphragm chamber of the actuator. This closes the control valve. As a result, the effective length of the intake manifold is lengthened and the intake efficiency in the low-to-medium speed range is improved due to the dynamic effect of the intake air, thereby increasing the power output.

: Effective Intake Manifold Length



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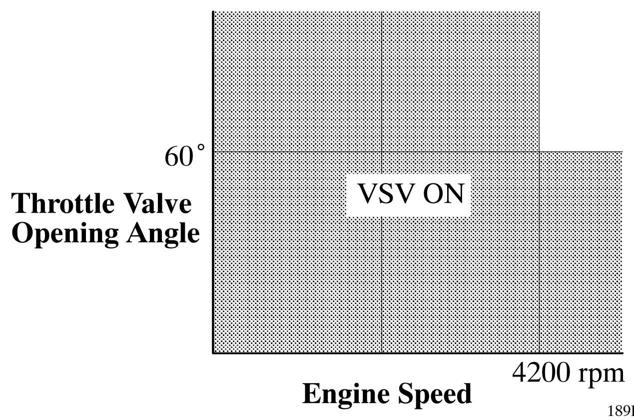

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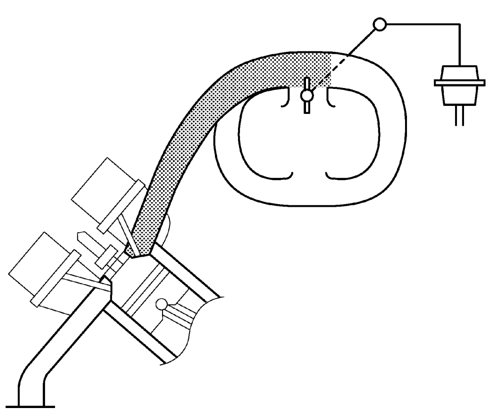
2) Intake Air Control Valve Open (VSV OFF)

The ECM deactivates the VSV to match the shorter pulsation cycle so that atmospheric air is led into the diagram chamber of the actuator. This opens the control valve. When the control valve is open, the effective length of the intake air chamber is shortened and peak intake efficiency is shifted to the high engine speed range, thus providing greater output at high engine speeds.

: Effective Intake Manifold Length



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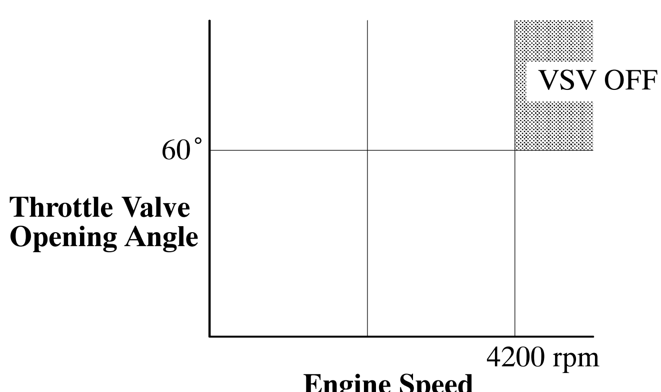

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9. Air Injection System (Only for European, Australian and Chinese Models)

General

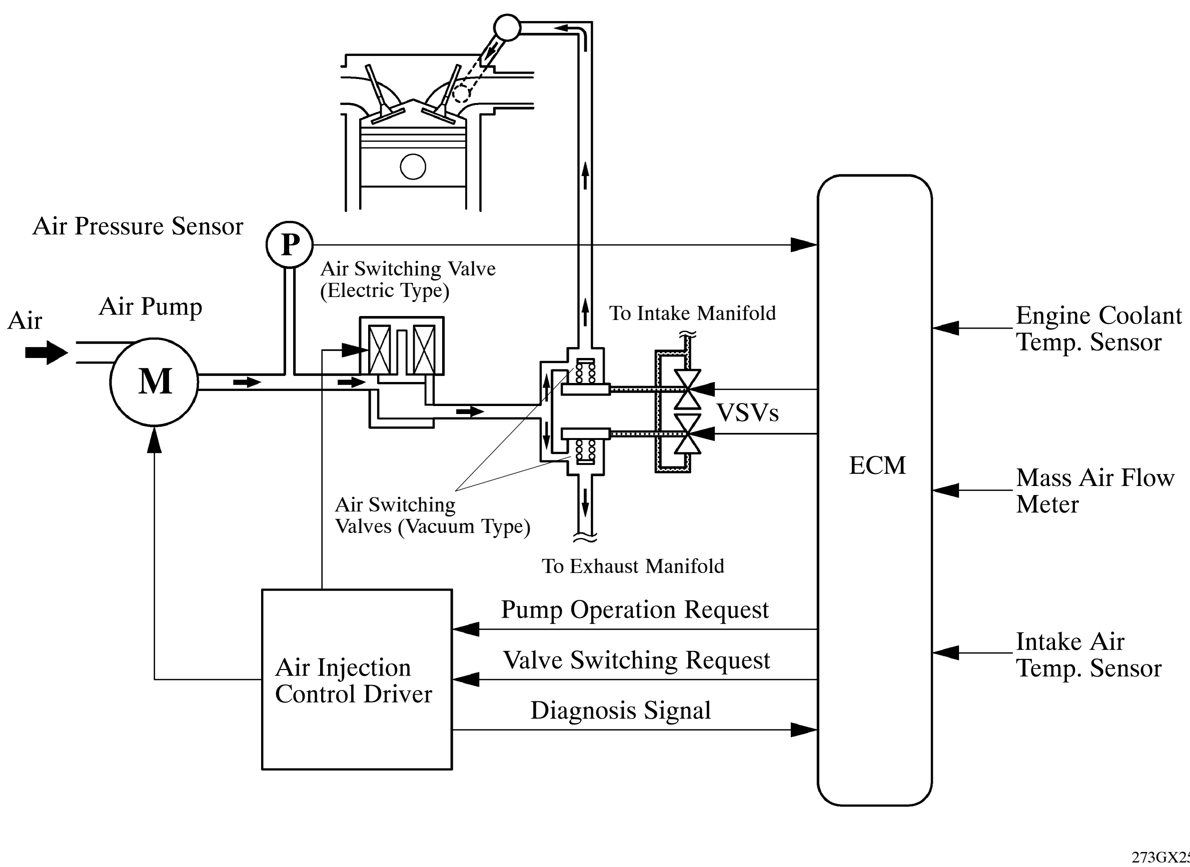
To ensure the proper warm-up performance of the TWC (Three-Way catalytic Converter) when starting a cold engine, an air injection system is used.

- This system consists of an air pump, an air injection control driver, two air switching valves (vacuum type), an air switching valve (electric type), and two VSVs.
- The ECM estimates the amount of the air injected to the TWC based on the signals from the mass air flow meter in order to regulate the air injection time.
- Air is injected under the following conditions.

▶ Operation Conditions ◀

Engine Coolant Temp.	5 to 45 °C (41 to 113 °F)
Intake Air Temp.	5 °C (41 °F) or more

▶ System Diagram ◀

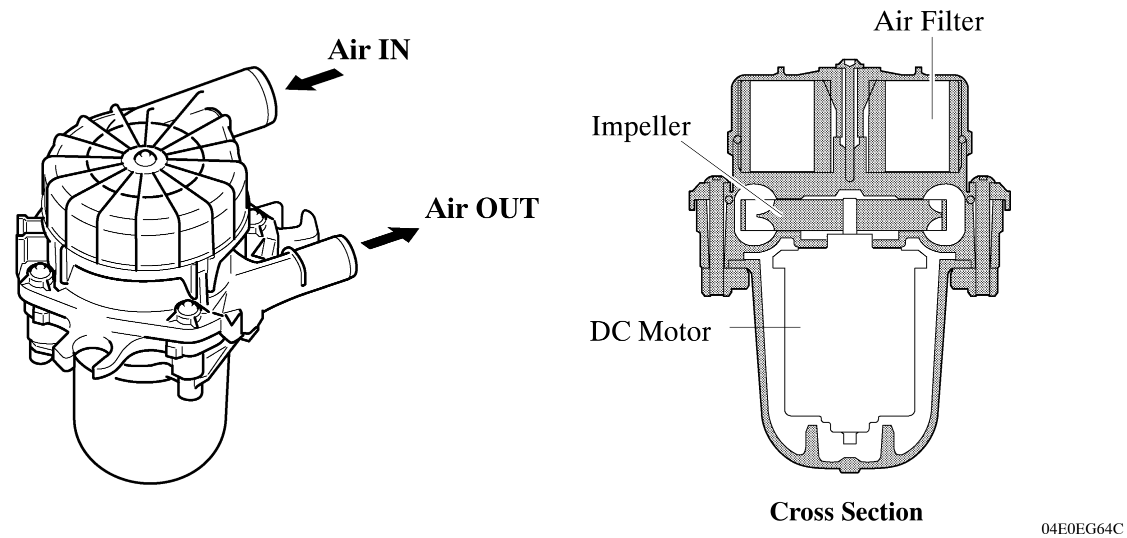


Construction and Operation

1) Air Pump

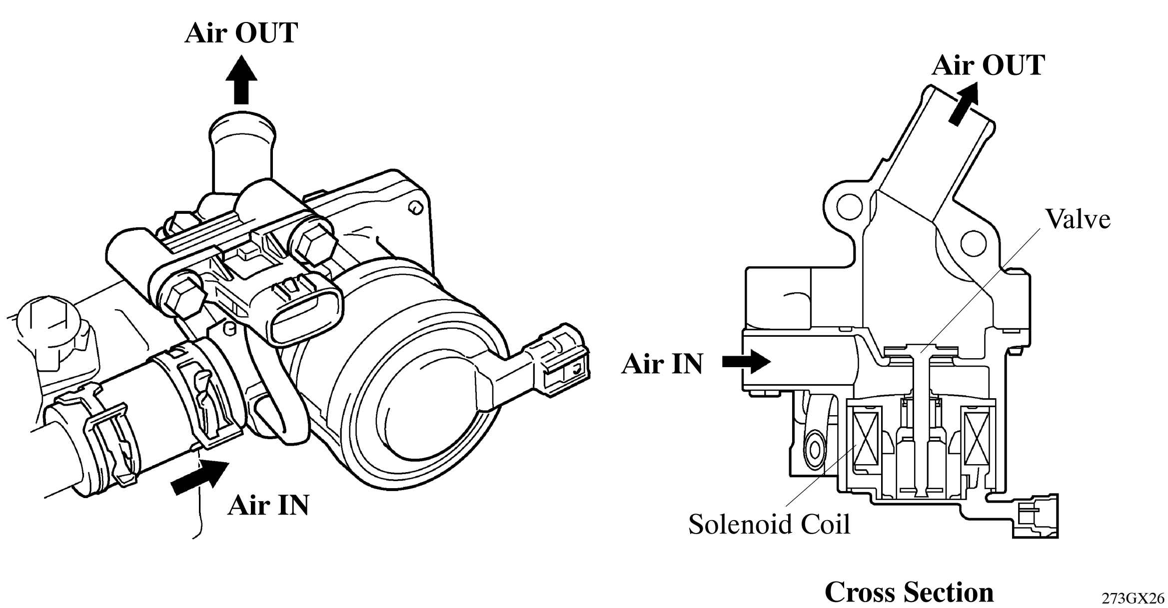
An air pump consists of a DC motor, an impeller and an air filter.

- The pump supplies air into an air switching valve (electric type) using its impeller.
- The air filter is maintenance-free.



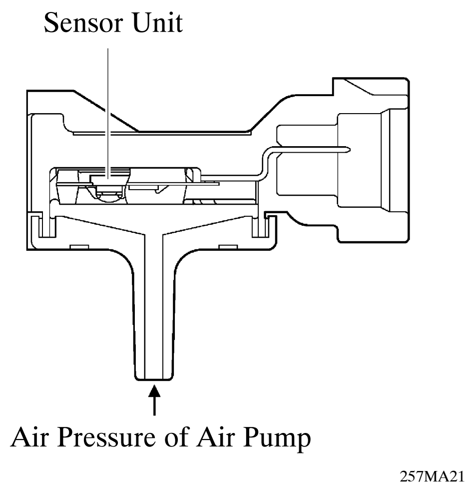
2) Air Switching Valve (Electric Type)

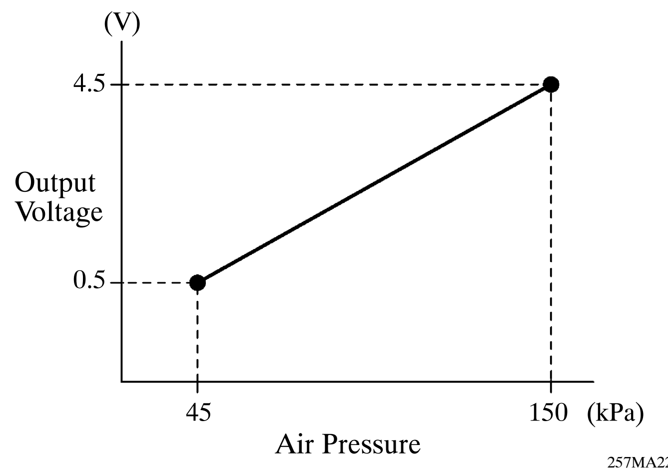
The air switching valve (electric type) is operated by a solenoid coil to control air injection and prevent back-flow of exhaust gas. Opening timing of the valve is synchronized with operation timing of the electric air pump.



3) Air Pressure Sensor

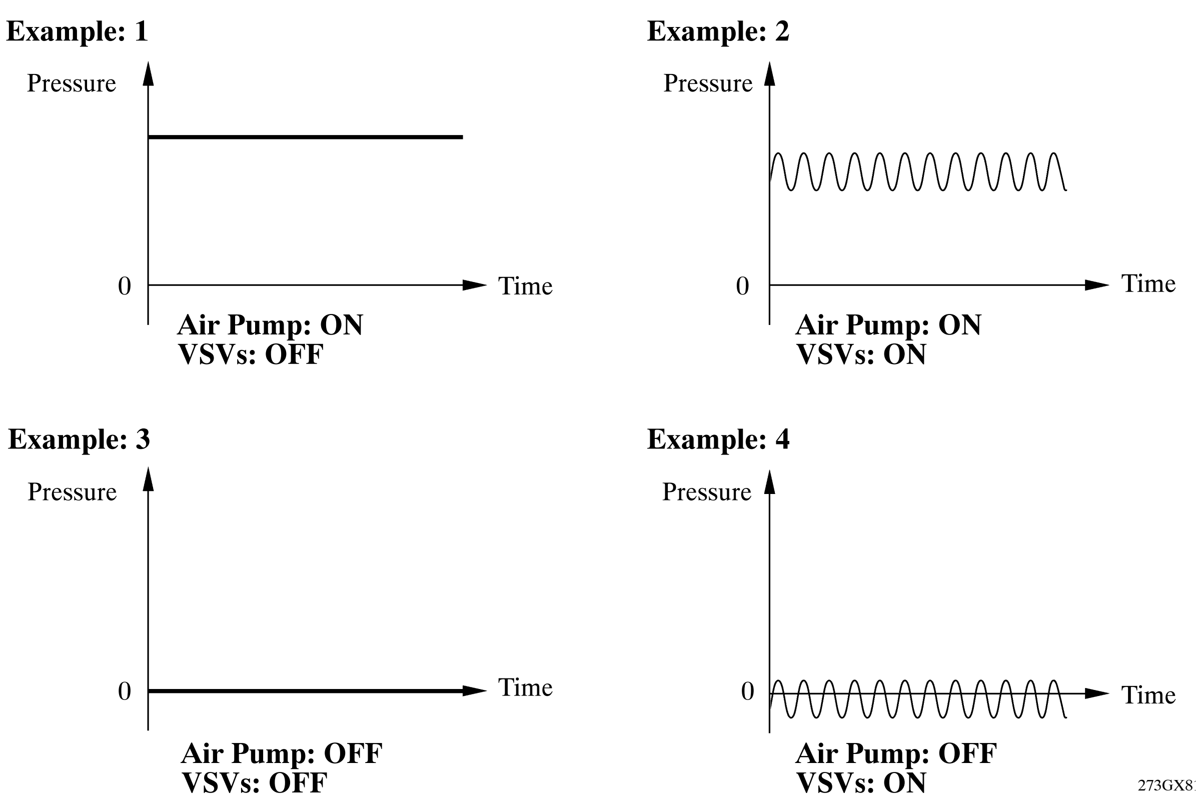
The air pressure sensor consists of a semiconductor, which utilizes the characteristics of silicon chip that changes its electrical resistance when pressure is applied to it. The sensor converts the pressure into an electrical signal, and sends it to the ECM in an amplified form.





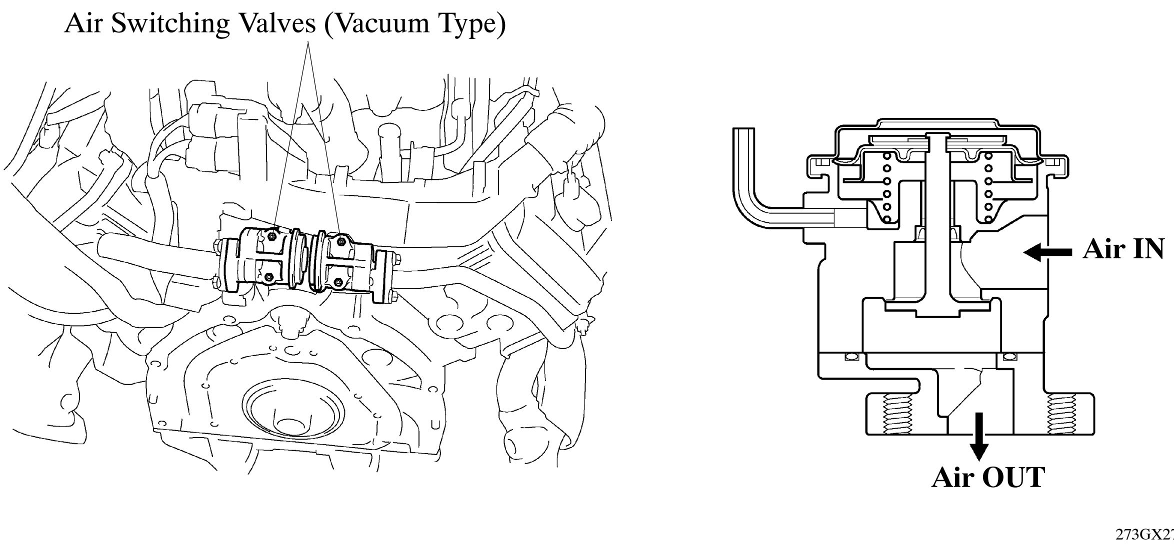
The ECM detects operation condition of the air injection system based on the signals from the air pressure sensor as follows:

1) When the air pump is ON and the VSVs are OFF, the pressure is constant.
2) When the VSVs are turned ON with the air pump ON and all air switching valves open, the pressure drops slightly and becomes unstable because of exhaust-pulse.
3) When the air pump is OFF and the VSVs are OFF, the pressure remains 0.
4) When the VSVs are turned ON with the air pump OFF and all air switching valves open, the pressure drops below 0 and becomes unstable because of exhaust-pulse.



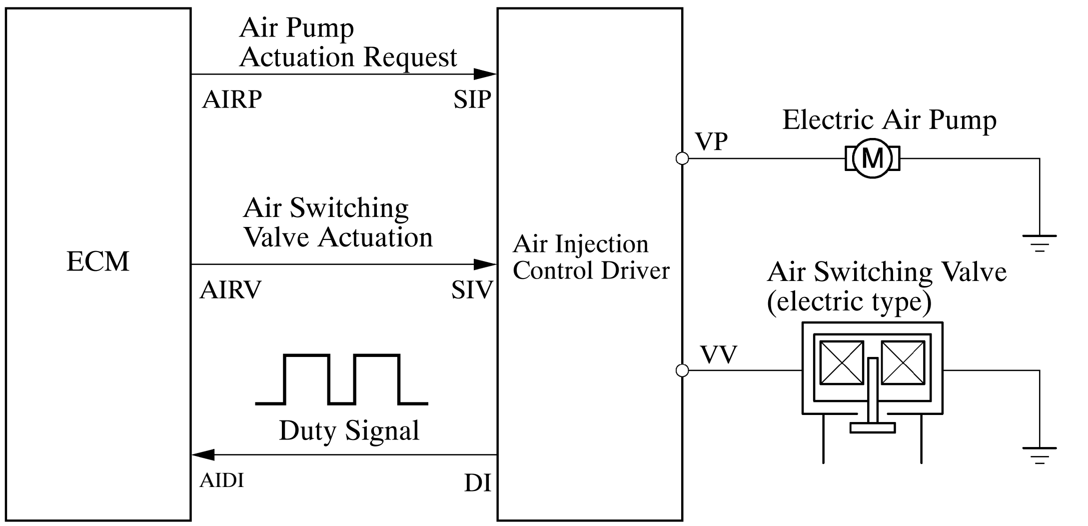
4) Air Switching Valve (Vacuum Type)

The air switching valve (vacuum type) switches according to the vacuum pressure introduced by VSV.



5) Air Injection Control Driver

- A semiconductor type air injection control driver is used. Activated by the ECM, this driver actuates the air pump and the air switching valve (electric type).
- The air injection control driver also detects failures in the input and output circuits of the air injection driver and transmits the failure status to the ECM via duty cycle signals.



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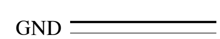
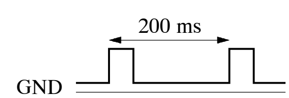
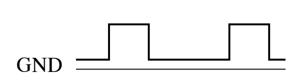
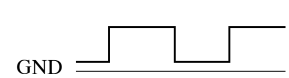
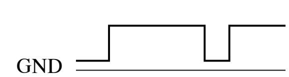
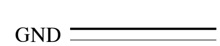
▶ DI Terminal Output ◀

Condition	AIRP	AIRV	Output (Duty Signal)
Open circuit in line between AIDI and DI terminals.	—	—	 273GX28
Failure in line between ECM terminals and air injection control driver.	—	—	 273GX29
Output failure at air injection control driver. (Failure in air pump actuation circuit)	—	—	 273GX30
Output failure at air injection control driver. (Failure in air switching valve actuation circuit)	—	—	 273GX31
Overheat failure of air injection control driver.	—	—	 273GX32
Normal	ON	ON	 273GX33
	OFF	OFF	 273GX29
	ON	OFF	
	OFF	ON	

10. Diagnosis

- When the ECM detects a malfunction, the ECM makes a diagnosis and memorizes the failed section. Furthermore, the check engine warning light in the combination meter illuminates or blinks to inform the driver.
- The ECM will also store the DTC (Diagnostic Trouble Code) of the malfunctions. The DTC can be accessed by using the intelligent tester II.
- For details, refer to the Land Cruiser (Station Wagon) Repair Manual (Pub. No. RM0810E).

Service Tip

To clear the DTC that is stored in the ECM, use the intelligent tester II, disconnect the battery terminal or remove the EFI MAIN fuse and ETCS fuse for 1 minute or longer.

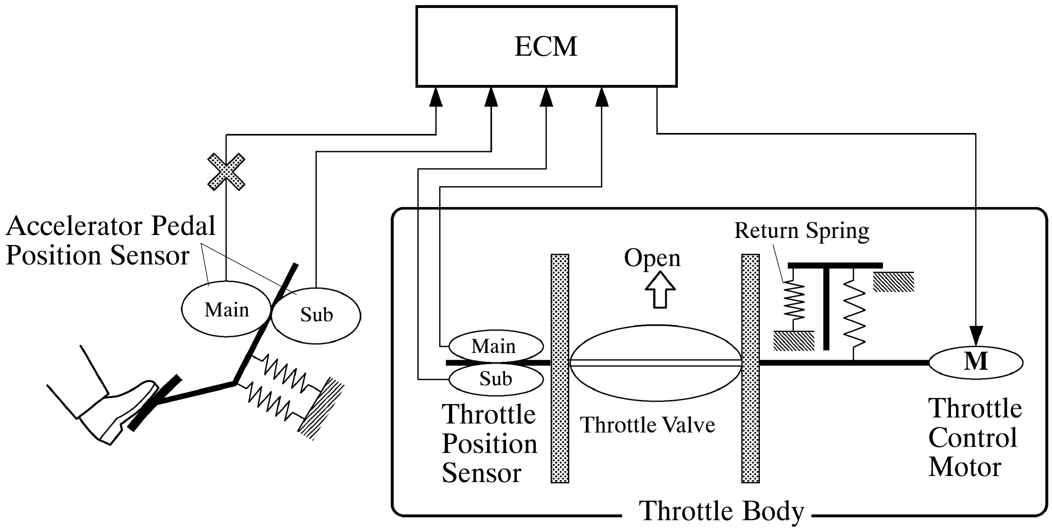
11. Fail-safe

General

When a malfunction is detected at any of the sensors, there is a possibility of an engine or other malfunction occurring if the ECM were to continue to control the engine control system in the normal way. To prevent such a problem, the fail-safe function of the ECM either relies on the data stored in memory to allow the engine control system to continue operating, or stops the engine if a hazard is anticipated. For details, refer to the Land Cruiser (Station Wagon) Repair Manual (Pub. No. RM0810E).

Fail-safe Operation due to Accelerator Pedal Position Sensor Trouble

- The accelerator pedal position sensor comprises two (Main, Sub) sensor circuits.
- If a malfunction occurs in either of the sensor circuits, the ECM detects the abnormal signal voltage difference between these two sensor circuits and switches into a fail-safe mode. In this fail-safe mode, the remaining circuit is used to calculate the accelerator pedal opening, in order to operate the vehicle under fail-safe mode control.



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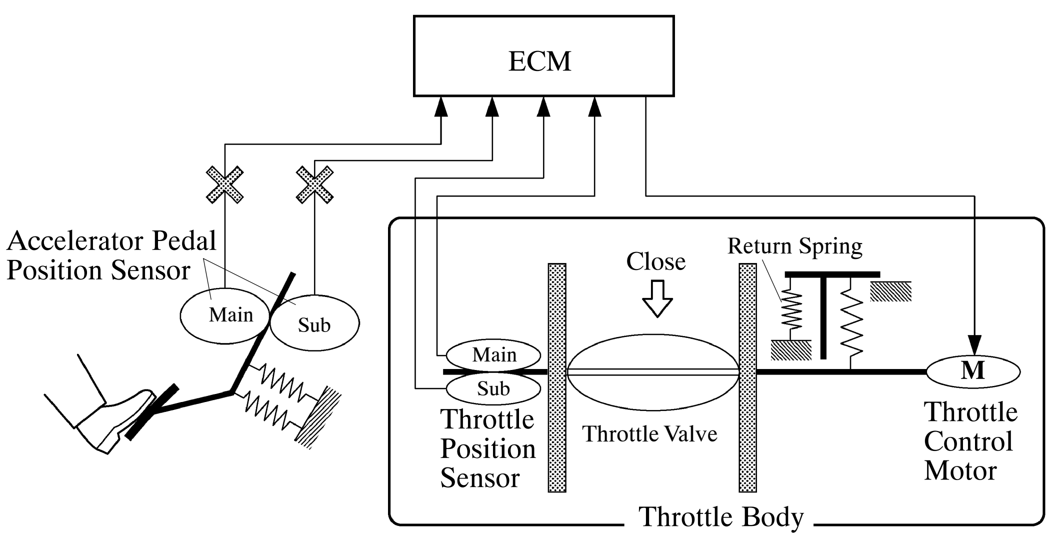
- If both circuits malfunction, the ECM detects the abnormal signal voltage from these two sensor circuits and discontinues the throttle control. At this time, the vehicle can be driven within its idling range.



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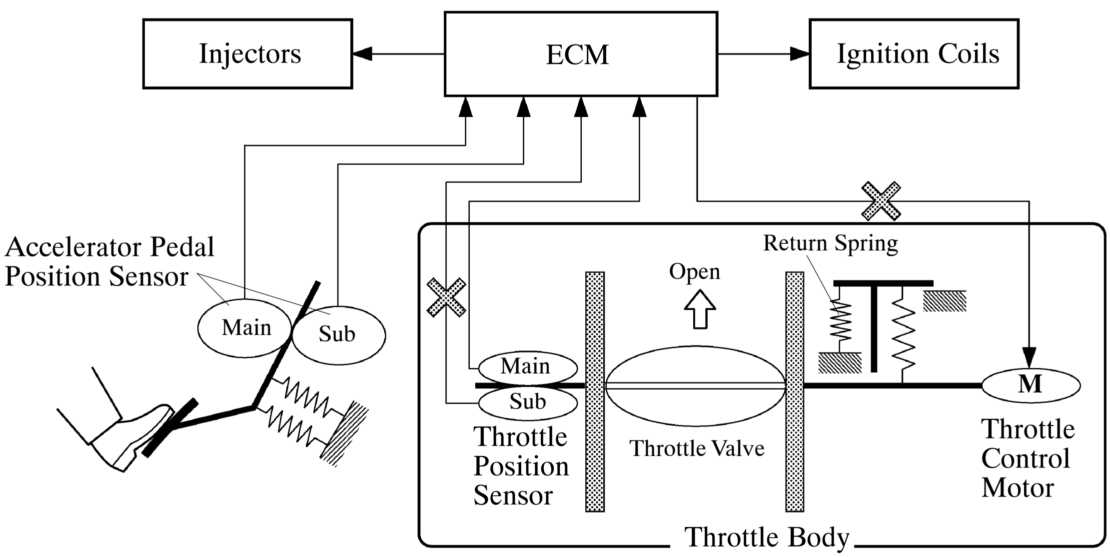
Fail-safe Operation due to Throttle Position Sensor Trouble

- The throttle position sensor comprises two (Main, Sub) sensor circuits.
- If a malfunction occurs in either of the sensor circuits, the ECM detects the abnormal signal voltage difference between these two sensor circuits, cuts off the current to the throttle control motor, and switches to a fail-safe mode.
- Then, the force of the return spring causes the throttle valve to return and stay at the prescribed base opening position. At this time, the vehicle can be driven in the fail-safe mode while the engine output is regulated through control of the fuel injection and ignition timing in accordance with the accelerator pedal position.
- The same control as above is effected if the ECM detects a malfunction in the throttle control motor system.



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